

Workflow Case 3

Joana Maria Nikita Dwicahyanti Nugroho

A. Objective

- a. Determine whether the individual carries a pathogenic variant associated with hereditary breast cancer risk.

B. Workflow

a. File Preparation

Mdkir capstone

b. Data Quality

Tools Used: Fastqc

Command:

Step 1: Quality Assessment with FastQC

```
cd capstone
```

```
cd case_3
```

```
conda activate 1.readq
```

```
fastqc case3_R1.fastq.gz
```

```
fastqc case3_R1.fastq.gz
```

Step 2: Trimming Reads with Trimmomatic

```
trimmomatic PE \
```

```
case3_R1.fastq.gz case3_R2.fastq.gz \
```

```
case3_R1_paired.fastq.gz case3_R1_unpaired.fastq.gz \
```

```
case3_R2_paired.fastq.gz case3_R2_unpaired.fastq.gz \
```

```
ILLUMINACLIP:TruSeq3-SE.fa:2:30:10:2:True
```

```
LEADING:30 TRAILING:30 MINLEN:36 HEADCROP:15 TAILCROP:5
```

Adapater yang digunakan adalah TruSeq3-SE. Trimmomatic menggunakan mode Pair-End.

Digunakan untuk menghapus sekuens adapter Nextera berdasarkan file adapter yang disediakan.

2 → jumlah maksimum mismatch yang diperbolehkan

30 → ambang batas palindrome clipping

10 → ambang batas simple clipping

2 → panjang minimum adapter untuk memicu proses penghapusan

True → mempertahankan kedua read meskipun hanya satu yang terkontaminasi adapter

LEADING:30 → Menghapus basa di awal setiap read jika memiliki nilai kualitas di bawah 30.

TRAILING:30 → Menghapus basa di akhir setiap read jika memiliki nilai kualitas di bawah 30.

HEADCROP:15→Menghapus 15 basa pertama dari setiap read tanpa mempertimbangkan kualitas.

TAILCROP:5→Menghapus 5 basa terakhir dari setiap read.

MINLEN:36→Menghapus read yang panjangnya kurang dari 36 basa setelah proses trimming, untuk memastikan hasil alignment tetap akurat dan andal.

Step 3: Quality Assessment after Trimming

```
fastqc case3_R1_paired.fastq.gz case3_R1_unpaired.fastq.gz
```

```
fastqc case3_R2_paired.fastq.gz case3_R2_unpaired.fastq.gz
```

c. Read Alignment

Tools: BWA MEM

Command:

Step 1: Read Alignment

```
bwa mem -t 1 \
```

```
-R '@RG\tID:a1\tSM:nor\tPL:ILLUMINA\tLB:lib1\tPU:f_1' \
```

```
~/capstone/reference/Homo_sapiens_assembly38.fasta \
```

```
~/capstone/case_3/case3_R1_paired.fastq.gz \
```

```
~/capstone/case_3/case3_R2_paired.fastq.gz \
```

```
> ~/capstone/outfile/sample_case3.sam
```

Step 2: Converting sam to bam

```
samtools view -Shb \
```

```
-o ~/capstone/outfile/sample_case3.bam \
```

```
~/capstone/outfile/sample_case3.sam
```

Step 3: Sort BAM File

```
samtools sort \
```

```
-o ~/capstone/outfile/sample3-sorted.bam \
```

```
~/capstone/outfile/germline.bam
```

Step 4: Index BAM File

```
samtools index \
```

```
~/capstone/outfile/sample_case3-sorted.bam
```

Step 5: Mark Duplicate Reads

```
picard MarkDuplicates \
```

```
INPUT=$HOME/capstone/outfile/sample_case3-sorted.bam \
```

```
METRICS_FILE=$HOME/capstone/outfile/sample_case3-met.txt \
```

```
OUTPUT=$HOME/capstone/outfile/sample_case3-dedup.bam \
```

```
CREATE_INDEX=true
```

Step 6: Base Quality Score Recalibration BQSR)

```
gatk BaseRecalibrator \
```

```
-R $HOME/capstone/reference/Homo_sapiens_assembly38.fasta \
```

```
-I $HOME/capstone/outfile/sample_case3-dedup.bam \
```

```
--known-sites
$HOME/capstone/reference/Homo_sapiens_assembly38.known_indels.vcf.gz \
-O $HOME/capstone/outfile/sample_case3.grp
```

Step 7: Apply BQSR

```
gatk ApplyBQSR -R ~/capstone/reference/Homo_sapiens_assembly38.fasta \
-I ~/capstone/outfile/sample_case3-dedup.bam \
-bqsr ~/capstone/outfile/sample_case3.grp \
-O ~/capstone/outfile/sample_case3-bqsr.bam
```

Step 8: Post Alignment QC

Untuk memunculkan analisis terkait dengan proses alignment.

```
gatk CollectAlignmentSummaryMetrics \
> -R ~/capstone/reference/Homo_sapiens_assembly38.fasta \
> --INPUT ~/capstone/outfile/sample_case3-bqsr.bam \
> --OUTPUT ~/capstone/outfile/sample_case3-
bqsr_Alignment_Summary2.txt

gatk CollectInsertSizeMetrics \
--INPUT ~/capstone/outfile/sample_case3-bqsr.bam \
--OUTPUT ~/capstone/outfile/sample_case3-bqsr_Size_Metrics.txt \
--Histogram_FILE ~/capstone/outfile/sample_case3-bqsr_Size_Histo.pdf
```

d. Variant Calling and Annotating

Step 1: Activate Conda

```
cd capstone
cd reference
conda activate 3.varcall
```

Step 2: Germline Variant Calling

```
gatk HaplotypeCaller \
-L chr13 \
-L chr17 \
-R ~/capstone/reference/Homo_sapiens_assembly38.fasta \
-I ~/capstone/outfile/sample_case3-sorted.bam \
-O ~/capstone/outfile/sample_case3.vcf
```

Step 3: Variant Filtering

```
gatk VariantFiltration \
-R ~/capstone/reference/Homo_sapiens_assembly38.fasta \
-V ~/capstone/outfile/sample_case3.vcf \
-O ~/capstone/outfile/sample_case3_filtered.vcf \
--filter-name "QD2" --filter-expression "QD < 2.0" \
--filter-name "FS60" --filter-expression "FS > 60.0" \
--filter-name "MQ40" --filter-expression "MQ < 40.0"
```

QD→Quality by Depth, mengukur kualitas varian per jumlah read

FS→Fisher Strand Bias, mengukur bias antara strand forward dan reverse

MQ→Mapping Quality, mengukur keyakinan read yang dipetakan

Step 4: Extract the Passed Variant

```
bcftools view \  
-f PASS \  
-O z \  
-o ~/capstone/outfile/sample_case3_pass.vcf.gz \  
~/capstone/outfile/sample_case3_filtered.vcf
```

Step 5: Create Zip dan Tbi file

```
bgzip ~/capstone/outfile/sample3_filtered.vcf  
tabix -p vcf ~/capstone/outfile/sample_case3_filtered.vcf.gz
```

Step 6: Annotating Germline Variant

```
bcftools annotate \  
-a ~/capstone/reference/clinvar_chr.vcf.gz \  
-c ID,INFO \  
-h <(bcftools view -h ~/capstone/reference/clinvar_chr.vcf.gz | grep  
"^##INFO=") \  
~/capstone/outfile/sample3_filtered.vcf.gz \  
-Oz -o ~/capstone/outfile/sample_case3_filtered_annotated.vcf.gz
```

Step 7: Extract to tsv. File

```
bcftools query -f  
"%CHROM\t%POS\t%REF\t%ALT\t%INFO/ALLELEID\t%INFO/RS\t%IN  
FO/CLNHGVS\t%INFO/GENEINFO\t%INFO/MC\t%INFO/CLNSIG\t%IN  
FO/CLNDN\t%INFO/ONC\t%INFO/SCI\t%INFO/SCIDN\n"  
sample_case3_filtered_annotated.vcf.gz >  
sample_case3_filtered_annotated_extracted.tsv
```

Step 8: Filtering on the the Pathogenic Variant

```
bcftools view \  
-i 'INFO/CLNSIG = "Pathogenic" || INFO/CLNSIG = "Likely_pathogenic" \  
~/capstone/outfile/sample_case3_filtered_annotated.vcf.gz \  
-Oz -o ~/capstone/outfile/BRCA_case3_pathogenic.vcf.gz
```